

2.4.6

EE25BTECH11016 - Taraka Abhinav

Question:

If $\mathbf{a} = 5\hat{i} - \hat{j} - 3\hat{k}$ and $\mathbf{b} = \hat{i} + 3\hat{j} - 5\hat{k}$, then show that the vectors $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$ are perpendicular.

Solution:

Two vectors are perpendicular if their dot product is zero. We need to show that $(\mathbf{a} + \mathbf{b})^\top(\mathbf{a} - \mathbf{b}) = 0$.

First, we find the vectors $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$. The given vectors can be written in column form as:

$$\mathbf{a} = \begin{pmatrix} 5 \\ -1 \\ -3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ -5 \end{pmatrix} \quad (1)$$

Now, we calculate the sum and difference:

$$\mathbf{a} + \mathbf{b} = \begin{pmatrix} 5 + 1 \\ -1 + 3 \\ -3 - 5 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \\ -8 \end{pmatrix} \quad (2)$$

$$\mathbf{a} - \mathbf{b} = \begin{pmatrix} 5 - 1 \\ -1 - 3 \\ -3 - (-5) \end{pmatrix} = \begin{pmatrix} 4 \\ -4 \\ 2 \end{pmatrix} \quad (3)$$

Next, we compute the dot product of these two resultant vectors:

$$\begin{aligned} (\mathbf{a} + \mathbf{b})^\top(\mathbf{a} - \mathbf{b}) &= \begin{pmatrix} 6 & 2 & -8 \end{pmatrix} \begin{pmatrix} 4 \\ -4 \\ 2 \end{pmatrix} \\ &= (6)(4) + (2)(-4) + (-8)(2) \\ &= 24 - 8 - 16 \\ &= 0 \end{aligned} \quad (4)$$

Since the dot product is zero, the vectors $(\mathbf{a} + \mathbf{b})$ and $(\mathbf{a} - \mathbf{b})$ are perpendicular. Hence, proved.

3D Plot of Vectors and Endpoints

